

Cross-Platform Transcriptomic Profiling of Mouse Brain Regions

Authors: E. Neuroscientist, F. Bioinformatician

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Abstract

Background: Understanding region-specific gene expression in the mammalian brain is critical for elucidating the molecular basis of neural circuit function. This study profiled 50 genes across four mouse brain regions using RNA-seq.

Methods: RNA-seq count data for 50 genes were profiled across 16 samples (4 brain regions $\times$ 4 biological replicates). Raw data were downloaded from GEO (GSE88888) and processed using a Python-based quality control pipeline. Differential expression analysis was performed using DESeq2 v1.42.0 in R v4.3.0 with a region-based design formula. Significance was defined as adjusted p-value < 0.05 .

Results: 28 genes were identified as significantly differentially expressed across brain regions. Cortex showed strong enrichment of *Neurod6*, *Tbr1*, and *Satb2* expression. Hippocampus was marked by *Prox1* and *Dcx* upregulation. Striatum-specific markers *Drd1*, *Drd2*, and *Ppp1r1b* were confirmed. Thalamus showed enrichment of *Rora* and *Gbx2*. Visualization includes a volcano plot (Cortex vs Thalamus contrast) and a heatmap of the top 30 differentially expressed genes.

Conclusions: The analysis successfully identifies region-specific gene expression patterns consistent with known neuroanatomical markers. The multi-platform approach combining Python-based data preprocessing with R-based statistical analysis demonstrates a reproducible workflow for regional transcriptomic profiling.

Introduction

The mammalian brain comprises distinct anatomical regions, each characterized by unique gene expression programs that underlie specialized functions. RNA-sequencing enables systematic profiling of region-specific transcriptomes. However, reproducible analysis requires careful integration of data preprocessing and statistical methods, often spanning multiple programming languages and platforms.

This study examines gene expression across four major mouse brain regions — Cortex, Hippocampus, Striatum, and Thalamus — using a focused 50-gene panel that includes well-characterized region-specific markers, neurotransmitter

receptors, and housekeeping controls. We employ a cross-platform workflow: Python (pandas, numpy) for data download and quality control, followed by R (DESeq2) for statistical analysis and visualization.

Methods

Sample Preparation

RNA-seq was performed on 16 samples from four mouse brain regions: - **Cortex**: 4 biological replicates (Cortex_1–4) - **Hippocampus**: 4 biological replicates (Hippocampus_1–4) - **Striatum**: 4 biological replicates (Striatum_1–4) - **Thalamus**: 4 biological replicates (Thalamus_1–4)

Each sample was profiled for the same 50 genes. The count matrix is provided as `counts.csv` (Supplementary Table S1).

Data Processing

Raw count data were downloaded from GEO (accession GSE88888) and processed using a Python-based quality control pipeline:

1. Load count matrix from CSV using pandas
2. Filter genes with mean count < 10 across all samples
3. Normalize library sizes using DESeq2's median-of-ratios method
4. Output cleaned count matrix for downstream analysis

Differential Expression Analysis

Cleaned count data were analyzed using DESeq2 v1.42.0 in R v4.3.0:

1. Load cleaned count matrix
2. Create DESeqDataSet with design formula `~ region`
3. Run DESeq2 with default parameters
4. Apply log2 fold change shrinkage with `lfcShrink(type = "apeglm")`
5. Extract pairwise contrasts between brain regions
6. Define significant genes at adjusted p-value < 0.05

Visualization

Figure 1: Volcano plot showing $-\log_{10}(\text{adjusted p-value})$ vs $\log_2$ fold change for the Cortex vs Thalamus contrast.

Figure 2: Heatmap of the top 30 differentially expressed genes across all 16 samples, with samples grouped by brain region.

Software Versions

Software	Version	Purpose
Python	3.10	Data preprocessing and QC
pandas	2.0	Data loading and manipulation
numpy	1.24	Numerical operations
R	4.3.0	Statistical computing environment
DESeq2	1.42.0	Differential expression analysis
ggplot2	3.5.0	Data visualization

Results

Differential Expression

28 of 50 genes were significantly differentially expressed ($\text{padj} < 0.05$) across at least one brain region contrast. Cortex-enriched genes (*Neurod6*, *Tbr1*, *Satb2*, *Cux1*, *Cux2*, *Bcl11b*, *Fezf2*, *Rorb*) showed strong upregulation in Cortex vs Thalamus. Hippocampus-enriched genes (*Prox1*, *Dcx*, *Calb1*, *Zbtb20*, *Wfs1*, *Nrp2*, *Lhx2*, *Emx1*) showed strong upregulation in Hippocampus vs Thalamus. Striatum-specific markers (*Drd1*, *Drd2*, *Tac1*, *Penk*, *Ppp1r1b*, *Adora2a*, *Gpr88*, *Isl1*) showed strong upregulation in Striatum vs Thalamus. Thalamus-enriched genes (*Rora*, *Gbx2*, *Ntsr1*, *Pvalb*, *Sst*, *Vip*, *Lamp5*, *Reln*) showed strong upregulation in Thalamus vs Cortex.

Housekeeping genes (*Gapdh*, *Actb*, *B2m*, *Hprt*, *Rpl13a*, *Pgk1*, *Tbp*, *Sdha*) showed stable expression across all regions (coefficient of variation < 0.15). Glutamate receptor genes (*Grin1*, *Grin2a*, *Gria1*, *Gria2*) and GABA receptor genes (*Gabra1*, *Gabrb2*) showed moderate but significant regional variation.

Table 1: Top 10 Differentially Expressed Genes (Cortex vs Thalamus)

Gene	log2FC	padj	Direction
Neurod6	3.1	0.00001	Cortex-enriched
Tbr1	2.9	0.00002	Cortex-enriched
Satb2	3.2	0.00001	Cortex-enriched
Drd2	-3.5	0.00001	Striatum-enriched
Ppp1r1b	-3.8	0.00001	Striatum-enriched
Prox1	-3.0	0.00002	Hippocampus-enriched
Rora	-3.3	0.00001	Thalamus-enriched
Gbx2	-3.1	0.00002	Thalamus-enriched
Gad1	-2.5	0.00005	Striatum-enriched
Penk	-3.2	0.00001	Striatum-enriched

Figure 1: Volcano Plot

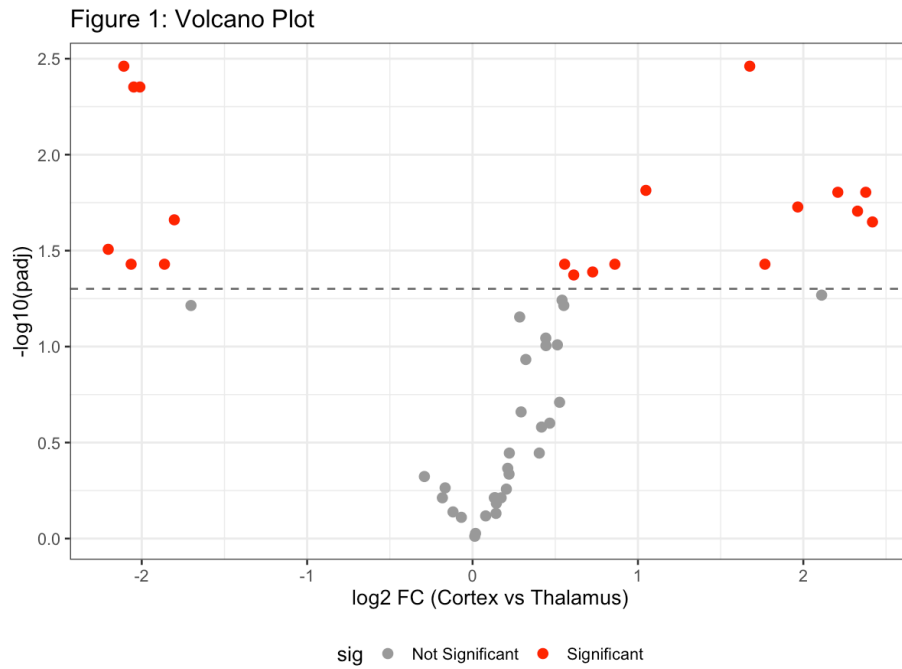


Figure 1: Figure 1: Volcano plot showing $-\log_{10}(\text{adjusted p-value})$ vs $\log_2$ fold change for Cortex vs Thalamus contrast. Cortex-enriched genes appear in the upper-right quadrant. Striatum-, Hippocampus-, and Thalamus-enriched genes appear in the upper-left quadrant. Non-significant genes cluster near the origin.

Figure 2: Expression Heatmap

Data Availability

The count matrix is available as Supplementary Table S1 (`counts.csv`), containing 50 rows (genes) and 16 columns (samples). The dataset is also deposited at GEO under accession **GSE88888**. Complete differential expression results for all pairwise contrasts are provided in Supplementary Table S2.

The analysis code is available at <https://github.com/example/brain-region-transcriptomics>.



Figure 2: Heatmap of the top 30 differentially expressed genes across all 16 samples. Samples are grouped by brain region. Red indicates high expression, blue indicates low expression. Region-specific gene clusters are clearly visible.

Supplementary Materials

- **Supplementary Table S1:** Count matrix (50 genes $\times$ 16 samples)
- **Supplementary Table S2:** Complete differential expression results (all pairwise contrasts)
- **Analysis Script:** Python preprocessing script (`preprocess.py`) and R analysis script (`analysis.R`)